How Low Can You Go? Video Resolution and the Reliability of Markerless Pose Estimation in Bouldering

Đăng Khôi Bùi

October 20, 2026

1 Introduction

Markerless pose estimation lets researchers study human movement, and the perceptual–motor processing behind it, from ordinary video instead of lab motion capture [1, 2]. Climbing is a good test case: joint kinematics are used to infer route reading, fluency and skill [3, 4]. But the footage in such studies is usually phone video of very variable quality, compressed, downscaled for storage, or shot from far away so that the whole wall is in frame. Benchmark accuracy for pose models is reported on curated datasets such as COCO and does not tell a researcher how much resolution they can lose before their kinematic measures become unreliable. Studies of human-pose networks show that accuracy depends on how many pixels a person occupies: small or low-resolution people are detected and localised worse [5, 6]. Distal joints (wrists, ankles) are the smallest and most often hidden body parts, so they should be hit first. In climbing these are exactly the joints that carry the information: hand and foot placements.

Research question. How much does MediaPipe pose-tracking accuracy on bouldering video degrade as resolution is reduced, and is the loss larger for distal joints (wrists, ankles) than for proximal joints (shoulders, hips)?

2 Methods

Hypotheses. H1: agreement with the full-resolution reference (PCK) falls monotonically as resolution falls. H2: the fall is steeper for distal joints than for proximal joints (a resolution × joint-group interaction).

Data. Six bouldering clips recorded on a phone in a climbing gym (one climber, problems graded 5Q–2Q, 30 fps). Native resolution was 1080×1920 for five clips and 2160×3840 (4K) for one. The first 300 frames (10 s) of each clip were used.

Procedure. Each clip was processed with the MediaPipe PoseLandmarker (`full` model, VIDEO mode), the same unmodified pipeline as in the exploratory notebook (`day1.ipynb`). We used six resolution levels: native (reference) and long side 1280, 854, 640, 426 and 320 px (about 720p down to 180p, area interpolation). Because no hand-labelled ground truth was available, the native-resolution output serves as the *reference*. Accuracy is therefore measured as *agreement with full-resolution tracking*, which is the practical question for someone who stores or records at lower quality.

Measures. For joint j in frame t, the normalised error is

$$e_{jt} = \frac{\|\hat{\mathbf{p}}_{jt}^{(r)} - \mathbf{p}_{jt}^{(\text{native})}\|}{\ell_{\text{torso}}}, \qquad \text{PCK@0.1} = \frac{1}{N}\sum_{j,t} \mathbf{1}\left[e_{jt} < 0.1\right],$$

where ℓ_{torso} is the median shoulder–hip distance in the reference, so errors are comparable across camera distances. Frames where the person was missed at the lower resolution count as incorrect. Joints were grouped as distal (wrists, ankles), middle (elbows, knees) and proximal (shoulders, hips).

Statistical analysis. H1: for each joint group, a Friedman test across the five levels (clips as blocks), and Page's L trend test, which specifically tests the *ordered* alternative that accuracy declines with resolution. H2: a linear mixed model on log median error, $\log_{10} e_{gcr} = \beta_0 + \beta_1 \log_2 r + \beta_2\,\text{group} + \beta_3(\log_2 r \times \text{group}) + u_c + \varepsilon$, with a random intercept u_c per clip. Proximal is the reference group. H2 predicts a more negative slope for distal joints ($\beta_3 < 0$).

3 Results

H1 (monotonic decline): supported. Agreement fell as resolution fell for every joint group (Table 1, Fig. 1). Friedman tests: distal $\chi^2(4) = 23.8$, $p < .001$; middle $\chi^2(4) = 22.9$, $p < .001$; proximal $\chi^2(4) = 18.6$, $p < .001$. Page's trend test confirmed an ordered decline in every group (all $p < .001$). The decline was not gradual. Between 1280 and 640 px, distal PCK changed by only 3 points, but between 640 and 426 px it dropped by 14 points. In the same step the person was missed in 7% of frames instead of 1–2%. In the mixed model, each halving of resolution increased error by a factor of about $10^{0.247} \approx 1.8$ ($\hat{\beta}_1 = -0.247$ per doubling, SE 0.018, $z = -13.6$, $p < .001$).

Table 1: Mean agreement with native-resolution tracking across the six clips (300 frames each).

Long side (px)	1280	854	640	426	320
PCK@0.1, distal (wrists, ankles)	0.868	0.849	0.836	0.693	0.651
PCK@0.1, middle (elbows, knees)	0.909	0.890	0.877	0.749	0.707
PCK@0.1, proximal (shoulders, hips)	0.963	0.956	0.948	0.838	0.825
Median error, distal (torso lengths)	0.024	0.026	0.030	0.048	0.062
Median error, proximal (torso lengths)	0.012	0.014	0.017	0.025	0.034
Person detected (% of frames)	99.0	98.6	98.5	93.0	92.7

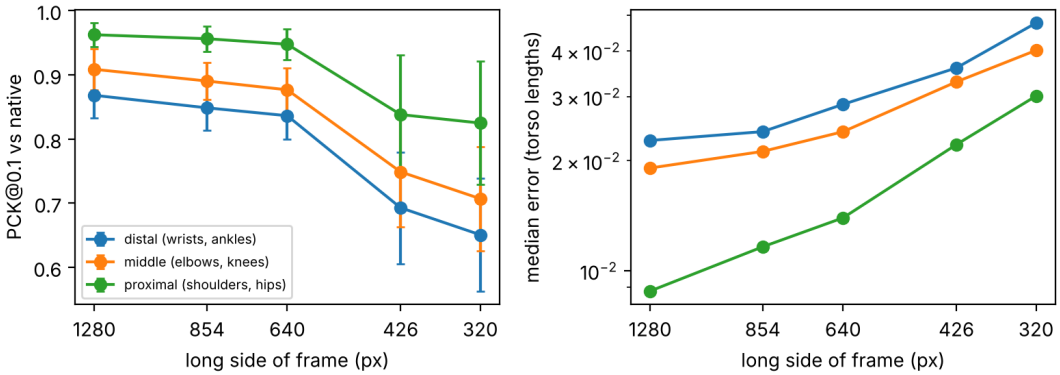


Figure 1: Left: PCK@0.1 against the native-resolution reference (mean ± SE over clips). Right: median error. The x-axis runs from high to low resolution.

H2 (distal joints hit harder): partly supported. In absolute terms distal joints lost more accuracy. The PCK drop from 1280 to 320 px was larger for distal than for proximal joints in all six clips (mean 0.217 vs. 0.138; one-sided Wilcoxon signed-rank $W = 21$, $p = .016$, the smallest p possible with $n = 6$). Distal error was also about twice the proximal error at every resolution. In *relative* terms, however, the interaction in the mixed model was not significant ($\hat{\beta}_3 = +0.040$, $p = .12$). Error grew by a similar *proportion* in every group, and distal joints lost more correct detections because they started closer to the 0.1-torso threshold.

4 Discussion

Interpretation. For bouldering video, MediaPipe tracking is robust down to about 640 px on the long side (roughly 360p in portrait), with less than 3 points of PCK loss compared with 1280 px. Below that there is a clear threshold: the person detector starts to miss frames, and hand and foot positions, the joints that carry route-reading and foot-placement information, lose agreement fastest in absolute terms. The model resizes its input to a small fixed size internally [1], so resolution matters mainly through the first stage, finding the person and cropping the region of interest. This explains why the loss appears suddenly instead of growing steadily. One clip (4Q Square D, climber far from the camera) lost more than 50 points even for proximal joints. The pixel size of the *person*, not of the frame, is probably what matters, which fits earlier findings on small-person pose estimation [5, 6].

Practical implication. Climbing studies that compute jerk or contact timing from phone video should keep the climber at least about 640 px tall in the frame. Below that, distal-joint kinematics should be filtered or excluded. Measures based on the hip or centre of mass hold up better.

Limitations and validity. (1) The reference is the model's own native-resolution output, not hand-labelled ground truth. The study measures *consistency* with full-quality tracking, and errors common to both resolutions are invisible. Labelling 50–100 frames would allow absolute accuracy to be measured. (2) Six clips and one climber, the first 10 s only. Clips are the unit of replication, so the tests have little power and $n = 6$ limits the Wilcoxon test's smallest possible p. (3) Resolution was reduced by downscaling, which differs from genuinely low-quality recording (compression, motion blur). (4) Frame-rate reduction, the second part of the original question, was not tested in the time available and is the natural next step, together with person-size in pixels as a continuous predictor.

References

[1] V. Bazarevsky et al. BlazePose: On-device real-time body pose tracking. *arXiv:2006.10204*, 2020.

[2] A. Mathis et al. DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. *Nature Neuroscience*, 21:1281–1289, 2018.

[3] L. Seifert et al. Climbing skill and complexity of climbing wall design: assessment of jerk as a novel indicator of performance fluency. *J. Applied Biomechanics*, 30(5):619–625, 2014.

[4] P. Cordier, M. Mendès France, J. Pailhous, P. Bolon. Entropy as a global variable of the learning process. *Human Movement Science*, 13:745–763, 1994.

[5] T.-Y. Lin et al. Microsoft COCO: Common objects in context. *ECCV*, 2014.

[6] K. Sun, B. Xiao, D. Liu, J. Wang. Deep high-resolution representation learning for human pose estimation. *CVPR*, 2019.